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Linking Psychophysiological Markers To Situational Performance: An EEG Study of Police Cadets during Critical Incident Simulations

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Abstract

Physiological measures, most commonly heart rate, are widely used in applied police research to assess the relationships between situational stress and officer performance under pressure. However, measurements of the neurocognitive mechanisms underlying these critical skills remain limited, especially throughout police academy training. This study investigates the potential of electroencephalography (EEG) and its relationship to situational performance outcomes in police cadets ($n = 58$) at Kuwait's National Police Academy. EEG and electrocardiogram (ECG) activity were recorded as cadets from three different cohorts participated in a video simulation of a stressful critical incident, featuring seven decision prompts that called for procedural action. Cadets' decision-making, reasoning, and memory recall were rated during a post-task debriefing interview. Preliminary pairwise analyses identified significant correlations between performance metrics and neural activation in both beta and theta bands, particularly in the frontal cortex. Comprehensive multivariate analysis revealed frontal cortex beta-band activity to be a significant correlate of performance, particularly during decision-making and memory recall, underscoring its role in executive functions crucial to situational performance in policing. Contrary to studies that find higher activation leads to better outcomes, lower beta-band activation correlated to better performance. Additionally, ECG showed minimal predictive value during multivariate testing. This marks the first time EEG and ECG measures have been integrated into a single model predicting performance in policing. These findings contribute novel insights into the psychophysiological study of police performance, highlighting important implications for enhancing training, evaluation, and research methodologies in applied law enforcement settings.

Keywords Policing · Situational performance · Situational stress · Electroencephalography (EEG) · Electrocardiogram (ECG)

Introduction

A substantial body of applied psychophysiological research demonstrates that affective stimuli—most notably stress—influence cognitive and motor functions critical to situational performance among police officers (Anderson et al., 2019; Di Nota et al., 2020; Di Nota & Huhta, 2019; Hope et al., 2016; McClure et al., 2020; Petersson et al., 2017). A synthesis of theoretical and empirical models, such as Kelley et al.' (2019) biopsychosocial (BPS) framework, suggests

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that situational performance is shaped by perceived threats and challenges, both internal and external, that trigger psychological and physiological responses. However, measures of physiological determinants collected are often recurring, limiting variability and predictive power when assessing physiological correlates to performance among police personnel. Alhough limited in number, recent studies have begun employing more comprehensive methods to assess determinants of police performance, notably electroencephalography (EEG) (Alexander et al., 2024; Johnson et al., 2014; Muñoz et al., 2020). This article aims to build on these studies by providing a practical and accessible framework for assessment that can be readily applied by practitioners and stakeholders, while also addressing gaps in the literature related to advancements in the assessment of neurophysiological markers and their potential correlations with situational performance in policing.

Contemporary studies investigating police performance primarily utilize virtual or live simulations that mimic stressful encounters with the public, focusing on identifying the critical skills, traits, and factors that impact police performance. Though often simplified as "shoot/no-shoot" choices, police officers make various complex decisions concerning their actions and commands, which must align with legal protocols in their jurisdiction (Di Nota et al., 2021). Situational awareness directly informs decision-making, as officers must comprehend their surroundings and exercise sound judgment on the next course of action. Maintaining situational awareness and regulating emotions under pressure is crucial for effectively navigating stressful police encounters. Officers are continuously required to weigh potential motor strategies and resulting outcomes based on immediate perceptions, reasoning, and best practices imparted through training and experience (Baldwin et al., 2022; Di Nota et al., 2023; Huhta et al., 2023). Studies have also investigated memory recall and encoding in policing contexts, a critical element to learning, in-the-moment performance, and post-incident judicial proceedings (for a review, see Di Nota et al., 2020). Hope et al. (2016) demonstrate that both physical and mental stress experienced during critical incidents result in attentional narrowing, which limits the scope of information being encoded and subsequently impacts officers' ability to recall details accurately. Scholars have also explored specific elements found in critical incidents as determinants of memory encoding, particularly the prevalence of life-threatening weapons. Hulse and Memon (2006) showed that scenarios performed in a firearms training simulator induced greater physiological arousal than other scenarios, culminating in officers providing more accurate recollections, albeit less complete ones. Additionally, research suggests emotional salience plays a role in modulating working memory processes, with task-relevant emotional stimuli enhancing cognitive updating and memory performance under specific environments and situations (Berger et al., 2017).

Experience and training have also promoted optimal situational performance among police cadets and seasoned officers (Biggs et al., 2023; Saus et al., 2006; Suss & Boulton, 2020; Suss & Ward, 2010). Experienced officers are often capable of shifting effective strategies through "adaptive flexibility" (Boulton & Cole, 2016), while novices tend to follow more linear modes of decision-making and reasoning that are less effective during dynamic and uncertain situations. Tashman et al. (2006) concluded that experienced officers often rely on long-term working memory during anticipatory pre-scenario decision-making under dynamic and challenging conditions, responding faster and more accurately than their novice counterparts. Studies have also shown that firearm training and experience results in better performance under stress, favoring those with expertise in the necessary physical techniques relative to less experienced counterparts (Biggs et al., 2023; Vickers & Lewinski, 2012). Huhta et al. (2022, 2023) recently conducted a multi-modal series of investigations that collected neurophysiological eye-tracking data and post-incident debrief interviews among experienced officers and first and third-year cadets at Finland's Police University College. Their findings show that experts have more effective visual scanning techniques and a more comprehensive understanding of relevant situational elements. The experts' ability to more keenly observe, understand, and reason about what they would do next was acquired through formal training and experience-induced learning. This current study will focus on investigating situational performance skills among cadets at Kuwait's National Police Academy, which has a four-year program and a one-year specialized program[1] for cadets with prior post-secondary education. For the first time, the current study will also track learning- and experience-induced changes to neurophysiological activity using electroencephalography (EEG) in applied police settings.

Stress physiology is most commonly measured in applied police settings via ambulatory electrocardiography (ECG) and reported as heart rate (HR) (Andersen & Gustafsberg, 2016; Bertilsson et al., 2020). Increasing in popularity, however, is the measurement of heart rate variability (HRV), which comprises numerous metrics serving as indicators of balance between the autonomic "flight or fight" response and the parasympathetic "rest and digest" response (for review, see Andersen et al., 2024). The release of hormones

[1] The specialized program, often consisting of older cadets, is commissioned to attract specialized degree holders (e.g., engineers, physicians, etc.) to fill roles in the police force requiring specialization. They receive accelerated learning that is more basic than the one's provided in the four-year program.

like cortisol and adrenaline during high-stress police encounters has also been attributed to sensory distortions (e.g., tunnel vision) and significantly predicts lethal force decision-making errors Artwohl, 2002; Klinger & Brunson, 2009; McClure et al., 2020; Sapolsky, 2004). These findings underscore the importance of understanding how police officers process and prioritize goal-relevant objectives while regulating situational stress common in their lines of work.

Given the significant lack of EEG studies in applied police settings, we briefly review other relevant literature to understand the relationships between different EEG frequency bands and aspects of human performance. Neurophysiologically, increased power in the alpha-band (8–12 Hz) reflects inhibition of cortical activity to efficiently channel neural resources to task-relevant areas, which demonstrate increased power in the beta-band (13–30 Hz) linked to increased mental effort (Cheron et al., 2016). Similarly, alpha-band power typically decreases during psychosocial stress, whereas beta-band power tends to increase (Vanhollebeke et al., 2022). Specifically, stress increases reliance on the "habitual" memory system rather than cognitive systems, which are primarily regulated by the hippocampus and prefrontal cortex (Schwabe, 2024).

Extensive research across multiple fields has also focused on the neurological differences between novices and experts, revealing that expertise is associated with unique cognitive and neural processing patterns. Hatfield et al.' (2004) study on psychomotor performance found significant results when examining cortical activation in specific brain regions between experts and novices, particularly in terms of neural efficiency. Novices showed higher beta and gamma power, while experts demonstrated lower activation, resulting from a lesser need for attentional demands and cognitive interference, results similar to those found by Hill and Schneider (2006). Consequently, Bilalić et al. (2012) found distinct neural activation patterns in experts during chess, particularly along the dorsal stream, which is primarily located in the temporal and parietal lobes. According to their results, experts exhibited significantly different neural processes during object recognition and pattern processing. Studies have also focused on neural band fluctuations and their association with cognitive processes in association with expertise. Denis et al. (2017) found beta-band desynchronization during anticipatory tasks to be significantly different based on expertise, indicating a more refined neural response based on visual cues relevant to actions taken, something similarly examined on Orgs et al. (2008) similarly found in their sample of professional and non-dancers. These results are relevant to the current study, which samples cadets with varying training durations, holding significantly different levels of expertise that can potentially contribute to studies indicating neural differences among novices and experts.

Among the few studies investigating EEG in police contexts, Muñoz et al. (2020) measured HRV and EEG during virtual reality (VR) firearm training tasks that increased in difficulty. Their findings showed significant differences in frontal theta power and HRV time and frequency parameters during easy versus hard task levels and increased HR during all active shooting tasks in comparison to resting states. Using a similar VR task with varying levels of threat and difficulty, Alexander et al. (2024) explored oscillatory neural signals originating from the anterior cingulate cortex between experienced police officers and matched controls. Their results showed significant increases in theta power prior to an inhibited response (i.e., 'no-shoot' decision), which was significantly higher among experts who also demonstrated shorter beta rebounds in threat-attack conditions. Johnson et al.' (2014) research sought to distinguish psychophysiological indices from an expert and novice participants during deadly force judgment simulations. After completing 27 deadly force simulation scenarios, physiological measures—EEG and HRV—were correlated with performance. The results showed physiological measures to have stronger predictive power for expert performance, explaining 72% of variance, while novice performance exhibited only 37%. Higher cognitive workload and lower engagement—the psychophysiological state of gathering information through sustained attention—were found to be positively correlated to desirable performance outcomes. Experts needed less time to collect information when compared to less trained individuals, something that can significantly affect potential outcomes during critical incident scenarios that require swift action. These findings demonstrate the high potential of physiological measures in the study of high-stress policing, a topic this study aims to reinforce and elaborate on.

The current cross-sectional study seeks to extend existing knowledge on the neurophysiological correlates of situational performance during stressful encounters. Previous studies have provided valuable insights through the common use of ECG measures (mainly HRV), an easily accessible form of biological data collection. However, more advanced forms of physiological data collection can help fill significant gaps in the literature related to the neurological responses exhibited during these encounters. To address this gap, both EEG and ECG data will be collected from three cohorts of police cadets at Kuwait's National Police Academy as they participate in a professionally produced video-based simulation task, specifically designed to induce situational stress and elicit distinct decisions informed by the cadets' academy training. Based on the extant literature reviewed above from various disciplines and fields, we hypothesize that both frontal lobe EEG activity and HRV will be significant correlates of situational performance and that significant neurophysiological differences will be observed between cohorts with more training and experience.

The findings are expected to provide novel insights into the neurophysiological mechanisms underlying situational performance during critical police incidents. The potential results can demonstrate how neurophysiological measures, which are rarely used, can act as significantly more reliable indicators of performance outcomes compared to common methods like ECG. Additionally, this study offers insights into a framework that future research on situational performance can adopt to utilize more advanced physiological data collection techniques, potentially leading to more comprehensive findings in the development of this field.

Methods

Participants

A total of 58 male cadets from Kuwait's National Police Academy participated in the study (mean age = 22.8, SD = 3.15), including 21 first year cadets that had completed 4 months of training at the time of the study (mean age = 20.2, SD = 1.41), 20 specialized cadets—degree holders—and had completed 6 months of training (mean age = 26.50, SD = 2.09), and 17 third year cadets that had completed 30 months of training (mean age = 21.56, SD = 0.86) (see Supplementary Table 1). It is important to note that this study took place at Kuwait's primary police academy, where only male cadets are trained. Female cadets are taught at a separate, secondary academy.

Procedures

Before data collection commenced, information sessions were held for each cohort to explain the study's details. Volunteering participants provided written consent and were assured that the study was conducted for academic purposes. They were also informed that their data would be anonymized and kept strictly confidential, meaning it could not be traced back to individual participants or shared with the Police Academy except in aggregate form. Additionally, participants were reassured that their involvement, performance, or decision to withdraw from the study would not affect their academic standing in any way. Random stratified sampling was used to divide cadets depending on cohort. Random numbers were assigned to each cadet in each cohort to determine the sequence of participation. The experiment was approved by the Ethics Review Committee (ERC) of Kuwait University (Ethics Review Code: KU-ENG-14-11-23, approved on November 11th, 2023).

Video Simulation Task

Two waiting rooms were designated: one for participants awaiting their turn and another for those who had completed the study. These arrangements were made to prevent information sharing about the video simulation task between participants. Participants were escorted to the testing room and seated in front of a computer monitor when their number was called. They were fitted with EEG and ECG electrodes to monitor physiological responses during the 10-minute video simulation task. This video was produced by a local Kuwaiti production company, based on a script written by the principal author, who is experienced in use-of-force research in Kuwait's National Police Force. The input regarding the scenario script, which depicts a domestic violence call turning into a hostage situation, received significant feedback from on-duty officers who had similar experiences. They provided valuable input on elements that needed adjustment for better immersion before filming began. The simulated encounter in the video was designed in a point-of-view (POV) format to immerse the participants as responding officers, featuring a continuous 4-minute interaction that varied in complexity and urgency (Fig. 1).

Before the simulated encounter began, a 3-minute video was presented to establish a resting baseline, which featured a POV of walking on a paved walkway along the Kuwaiti shoreline at night with bystanders in the far distance. Next, the main 4-minute simulated encounter started with an audio distress call from an individual seeking help due to a sibling's threatening behavior. On-screen, participants received typical verbal dispatch information similar to what officers might encounter in the field. The simulation then transitioned to the POV of an officer arriving at the scene, unfolding over seven segments, each marked by visual prompts requiring participants to make a decision within a 5-second countdown. It is worth noting that no interactive methods of decision-making were incorporated into the simulation. Instead, cadets were asked to internalize their decisions to discuss during a debriefing interview. This approach, which is heavily influenced by limitations in technology used, will be explained in a later section. To prevent potential changes to initially internalized decisions, participants were informed that the video contained procedural mistakes, which aimed to discourage them from relying on the procedures shown as correct. The task was intentionally designed to induce stress similar to that experienced in high-pressure police situations. The scenarios included were uniformly presented to all participating cadets in the same order.

Each of the seven decision-making segments ranged in complexity and urgency, challenging participants' situational assessment in various contexts (detailed and visualized in

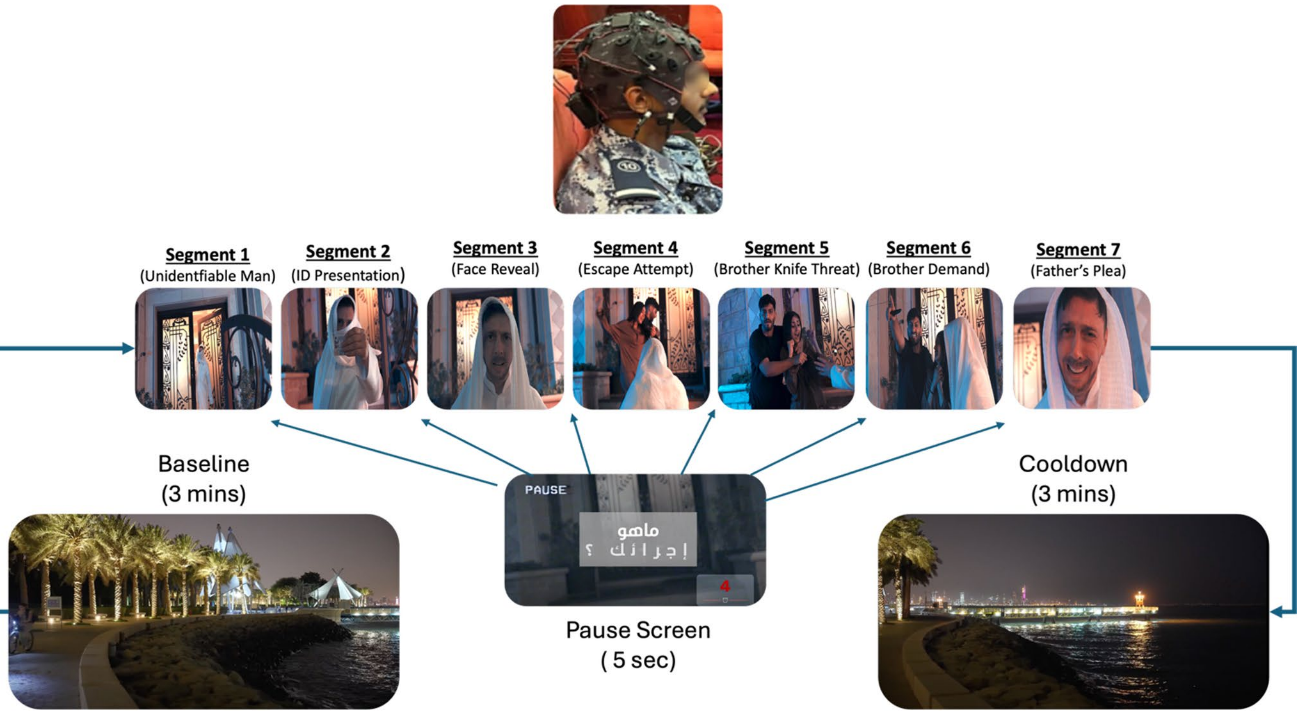

Fig. 1 Critical incident video simulation task. The image at the top shows a participating cadet wearing the EEG system. The collection of images below it show a breakdown of segments included in the video simulation task, including an initial 3-minute baseline video, 5-second decision prompts between seven segments, and final 3-min cooldown video

Fig. 1 below). After the conclusion of the video simulation task, the EEG and ECG electrodes were removed before directing participants to a designated room for a post-task interview and debrief (see Section "Situational Performance Data" for details).

EEG/ECG Collection

For the EEG recordings, a 32-channel EEG acquisition system (g.Nautilus Multi-Purpose; g.tec medical engineering GmbH, Schiedlberg, Austria) was utilized, with 31 channels designated for EEG signals and one channel allocated for ECG. The reference electrode was positioned on the left mastoid, and the ground electrode on the right mastoid. The electrode placement adhered to the standard 10–20 system, with the Oz electrode reassigned for ECG recording, positioned on the participant's left rib area.

As mandated by the police academy protocol, all cadets were required to have shaved heads, which facilitated easy electrode placement. Prior to fitting the EEG cap, the scalp and mastoid regions were cleansed using disinfectant wipes to enhance electrode contact. Dry electrodes were employed in this study instead of more conventional wet electrodes. Although wet electrodes are often regarded as the gold standard for EEG recordings, dry electrodes offer reliable performance (Hinrichs et al., 2020). The g.SAHARA dry electrodes used in this study are known for a lower signal-to-noise ratio compared to other dry-wireless EEG systems (Radüntz, 2018). Nonetheless, their straightforward setup and user-friendly design made them a practical choice, enabling efficient testing of approximately 20 cadets per day within the restricted 3-day study timeline.

Measures and Analyses

Situational Performance Data

Data collection for situational performance metrics involved one-on-one interviews conducted approximately 20 min after participants viewed the simulation, by the principal author who had developed the simulation task. Three distinct performance metrics were evaluated for each of the seven segments during the post-task interview: (1) decision making, (2) reasoning, and (3) memory recall. At the start of the interview, using a memory recall checklist (see Supplementary List 1), our interviewer asked participants to recall important events that had occurred during each segment of the simulation. Memory was scored ordinally (0 = no recall, 1 = partial memory, 2 = full memory) on recipients' ability to recollect 18 distinct events or elements of the simulation (for a maximum score of 36). Next, questions were asked about cadets' decision-making and the reasons behind decisions considered justified for each segment. These questions were open-ended and aimed to evaluate whether cadets understood the procedures and cues prompting their decisions. It should be noted that decision-making and reasoning scores were not based on the interviewer's discretion but were scored according to established use-of-force procedures currently mandated by the Kuwait National Police Force. The interviewer was responsible for determining whether these procedures or the reasoning behind them were correctly followed or considered—such as scanning for a weapon before using force or checking for visible signs of an altercation before proceeding with lawful entry. Reasoning and decision-making metrics were scored ordinally (0 = wrong, 1 = partially correct, 2 = correct) for a maximum score of 14, respectively. An example of potential answers and scores provided for one of the segments included in the simulation is shown in Fig. 2.

The decision to employ debriefing interviews instead of integrating interactive decision-making during the stimuli was due to technical difficulties with EEG technology, which is highly sensitive to movement, something that could introduce unwanted information during the collection process. Additionally, using options for interactive decision-making would not capture the nuances of critical incidents, especially the aspects of reasoning and memory. It should also be noted that the interviewer is neither employed by nor affiliated with Kuwait's police academy. Furthermore, the interviewer had no prior experience with any of the cadets interviewed and wore civilian clothing during these interviews to establish a neutral presence and promote stress-free, open conversations about the materials.

EEG/ECG Data Preprocessing

EEG data were recorded at 500 Hz and filtered using a Butterworth IIR bandpass filter between 1 and 40 Hz. The data were cleaned using EEGLab's (version 2024.0) automated artifact rejection to remove flat channels (minimum

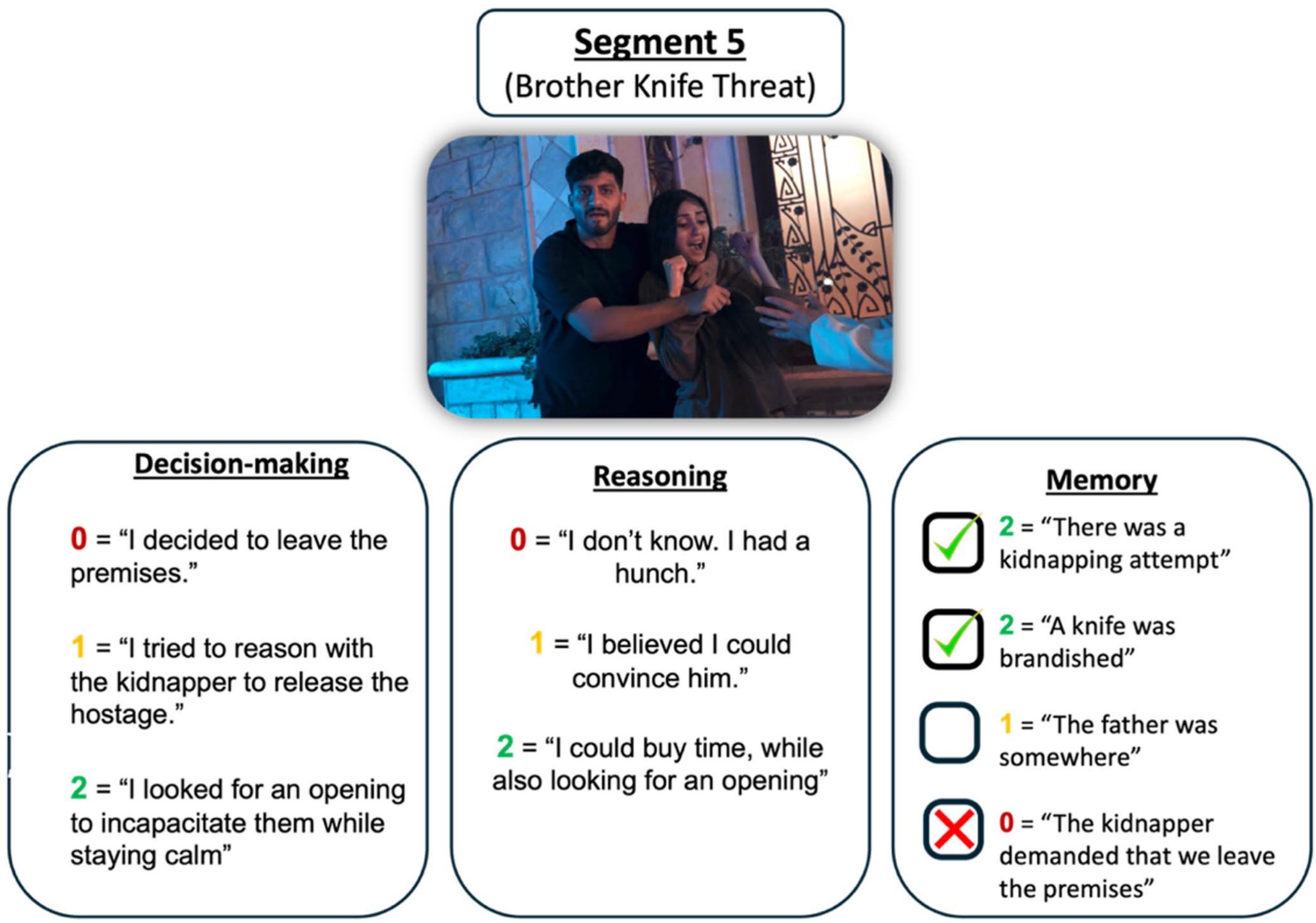

Fig. 2 Scoring example. After completing the video simulation task, each cadet participated in a debriefing interview. During this session, cadets were evaluated and scored on three situational performance metrics: decision-making, reasoning, and memory recall. Illustrated are examples of a participant's potential answers for segment 5, denoting what would warrant full (score of 2), partial (score of 1), and no points (score of 0) for decision-making and reasoning, respectively. The memory checklist on the bottom right would indicate how many of the 18 target items or elements of interest were freely recalled by the participant (score of 2), partially recalled (score of 1), or not recalled (score of 0); this segment had four memory recall items associated with it

inter-channel correlation threshold: 0.8) and correct bursts with the Artifact Subspace Reconstruction method (maximum SD = 20 per 0.5-second window). Additionally, segments with artifacts in ≥25% of channels exceeding 12 SD were excluded. Independent component analysis (ICA) was subsequently employed to identify and remove artifacts related to eye blinks and facial movements. One participant was excluded from final analysis due to equipment failure during data collection.

The clean EEG signals were then z-scored across the entire recording, including baseline and cooldown periods, to capture neurophysiological metrics relevant to critical police incidents and ensure comparability across participants by controlling electrode impedance variations. Afterward, z-scored EEG channels were then bandpassed using Butterworth IIR filters with the following cutoff frequencies corresponding to four major frequency bands of interest: 1–4 Hz (Delta), 4–8 Hz (Theta), 8–12 Hz (Alpha), and 12–30 Hz (Beta). The Hilbert Transform was then applied to extract the instantaneous amplitude of the signal and then averaged across the entire video stimulus. Previous work have investigated the neural correlates and predictive power of averaged neural activity in the order of minutes in relation to naturalistic behaviors and stimuli (Alasfour et al., 2022; Alasfour & Gilja, 2024). Channels were grouped into four regional lobes—frontal, temporal, parietal, and occipital—combining both hemispheres. This follows standard EEG preprocessing practice (Luck, 2005, p. 249; Pizzagalli, 2007; Zhang & Kappenman, 2024) with mean power within each region being computed to facilitate region-of-interest (ROI) analyses. Cronbach's Alpha was computed to verify internal consistency, with all channel clustering demonstrating strong reliability (α values: frontal = 0.9153, temporal = 0.9128, parietal = 0.9056, occipital = 0.8158). ROI analysis would also aid in inferring what EEG cap coverage would be suitable for further applications of this work.

Simultaneous ECG data were collected at 500 Hz. R-peaks were identified to compute RR intervals, which were resampled at 4 Hz using cubic spline interpolation. To remove ultra-low-frequency artifacts, RR intervals were detrended using the smoothness priors algorithm, ensuring stationarity (Tarvainen et al., 2002). Frequency domain analysis calculated power in low-frequency (LF: 0.04–0.15 Hz, reflecting sympathetic and parasympathetic influences) and high-frequency (HF: 0.15–0.4 Hz), linked explicitly to parasympathetic modulation bands (Castaldo et al., 2015). The normalized HF component (HFnorm), calculated as the ratio of HF power to total power, served as a measure of parasympathetic activity and stress modulation (Alasfour et al., 2021).

Analytical Approach

As reported elsewhere (AlSabah et al., under review), situational performance metrics were compared between cohorts, but novel between-group comparisons of EEG and ECG data are presented below. Statistical tests comparing performance metrics used ANOVA and Kruskal-Wallis tests, depending on whether the performance data was normally distributed or not. For this study, neurophysiological measures were used to explore pairwise between-group differences in different brain lobes across the previously mentioned frequency bands: delta, theta, alpha, and beta bands.

Preliminary pairwise correlation testing utilized Spearman Rho's to investigate potential significant relationships between our 5 physiological variables– averaged neural activity in frontal, temporal, parietal, and occipital lobes and HRV–and performance metrics in the form of decision-making, reasoning, and memory recall.

To explore the predictive relationships between our outcome variables (decision-making, reasoning, and memory recall) and explanatory variables (average beta-band activity in the frontal, temporal, parietal, and occipital lobes—as well as HRV), we used hierarchical linear modeling (HLM) to compare models' goodness of fit. This approach allowed us to assess the effects of the introduced variables. The first level included only frontal beta activity; the second added the remaining lobes (parietal, temporal, and occipital); and the third incorporated HRV. It should be noted that prediagnostic assumptions were met for all multivariate models used, with outliers removed based on the identified Cook's D. This slightly altered final sample sizes across models, which are included in the analyses in this article's results section. Furthermore, issues of multicollinearity were accounted for, with the highest variation inflation factor (VIF = 5.18) indicating a slight concern about multicollinearity in models using EEG only. The models reported below only utilized beta band activity. Other bands were unable to provide significant models, and therefore, they were excluded from reporting.

Statistical analyses and data visualization were performed using MATLAB (version R2024a; MathWorks, Natick, MA, USA), STATA (Release 17, StataCorp LLC), and R (Core Team 2024, R Foundation for Statistical Computing). The significance level for all statistical tests was set at $p < 0.05$. Additionally, the in-between analysis between cohorts adhered to the Bonferroni to account for Type I errors (*pBonf* = 0.0167).

Results

HRV Differences Between Baseline and Video Stimulus

To evaluate the effect of the video stimulus on heart rate variability (HRV), high-frequency normalized (HFnorm) values were compared across the baseline, stimulus, and cooldown phases. Outliers were removed using the interquartile range (IQR) method, and normality was assessed using the Lilliefors test. Due to evidence of non-normality in at least one phase (cooldown, $p < .05$), a non-parametric approach was applied. A Kruskal-Wallis test revealed no statistically significant difference in HFnorm across the three phases, $\chi^2(2) = 2.43$, $p = .297$. This indicates that no significant difference in HFnorm was exhibited between baseline, stimulus, and cooldown.

Pairwise Comparisons Between Cadet Cohorts

Situational Performance Metrics

According to supplementary research (AlSabah, Under Review) significant differences were found between our cohorts when examining situational performance metrics—decision-making, reasoning, and memory recall. ANOVA results for decision-making showed significant differences ($F(2,56)=13.36$, $pBonf=0.000$), while Mann-Whitney U tests for reasoning ($H=14.36$, $pBonf=0.001$) and ANOVA for memory ($F(2,56)=5.89$, $pBonf=0.005$) reinforced these findings (see Supplementary Table 2).

Pairwise comparisons indicated that first-year cadets scored significantly lower across all measures compared to third-year cadets ($pBonf<0.008$). While specialized cadets scored significantly higher than first-year cadets on reasoning ($z=3.46$, $pBonf=0.001$), no significant differences were observed for decision-making or memory. Notably, third-year cadets outperformed specialized cadets in decision-making ($t=6.06$, $pBonf=0.004$), though no significant differences were found for reasoning or memory.

Physiological Differences

After removing outliers (2 subjects), a Kruskal-Wallis test indicates frontal lobe beta-band activation ($H(2)=6.68$, $p=0.0288$) to be significantly different among our three police cohorts (see Fig. 3A). It should be noted, however, that in-between analysis for frontal beta activation did not result in any significant differences. All other comparison tests (Fig. 3B, C, and D) were not significant in either group or between-group comparisons.

EEG Correlates To Performance

Initial analysis illustrates significant channel-wise correlation coefficients between beta-band activity and each performance measure. For memory, no individual channel shows a statistically significant correlation with beta-band activity, although a slight increase in the magnitude of correlation coefficients is observed in the frontal region. In contrast, statistically significant correlations are observed at individual channels for both decision-making and reasoning, predominantly located in the frontal and parietal regions (see Fig. 4).

Proceeding with lobe-specific groupings, initial exploratory analysis revealed significant correlations, especially in both beta- and theta-bands in the frontal cortex (see Supplementary Table 3); however, only the most robust correlations maintained significance after multiple comparison testing with beta-band activation serving as the most prominent. Initial beta-band activation in the frontal cortex was strongly correlated with all 3 situational performance metrics: memory ($pBonf=0.0029$, $\rho=-0.345$), decision-making ($pBonf=0.0056$, $\rho = -0.328$), and reasoning ($pBonf=0.0006$, $\rho = -0.386$)–see Fig. 5. Most noticeably, all three metrics negatively correlate with frontal beta activity, such that higher situational performance were linked to lower beta-band power.

In addition to pairwise correlation analysis of the overall sample, Spearman correlations were computed to examine the relationship between frontal beta activity and performance metrics within each cohort group (First-Year, Specialized, Third-Year). A significant negative correlation was found between frontal beta activity and decision-making performance among First-Year cadets ($pBonf=0.0142$, $\rho = -0.59$). This suggests that increased frontal beta power was associated with poorer decision-making in this group. No other significant associations passed the Bonferroni threshold for in-group correlations.

Physiological Correlates of Situational Performance

For decision-making, all three models showed frontal beta activity to be significant. First-level analysis shows that beta-band activation in the frontal lobe had a significant negative effect, with a coefficient of -3.02 ($p<0.001$) and an adjusted R^2 of 0.13. Second-level modeling (adj $R^2 = 0.14$), with All Lobes included, showed the frontal lobe still maintaining significance, with a coefficient of -3.45 ($p<0.05$). Our third model, introducing HRV as an additional predictor variable, still consistently showed the frontal lobe as significant with a coefficient of -5.02 ($p<0.05$) with a negative effect of beta-band activation on decision-making, albeit in a non-significant model. Parietal and occipital lobes had negative but nonsignificant effects, while the temporal lobe

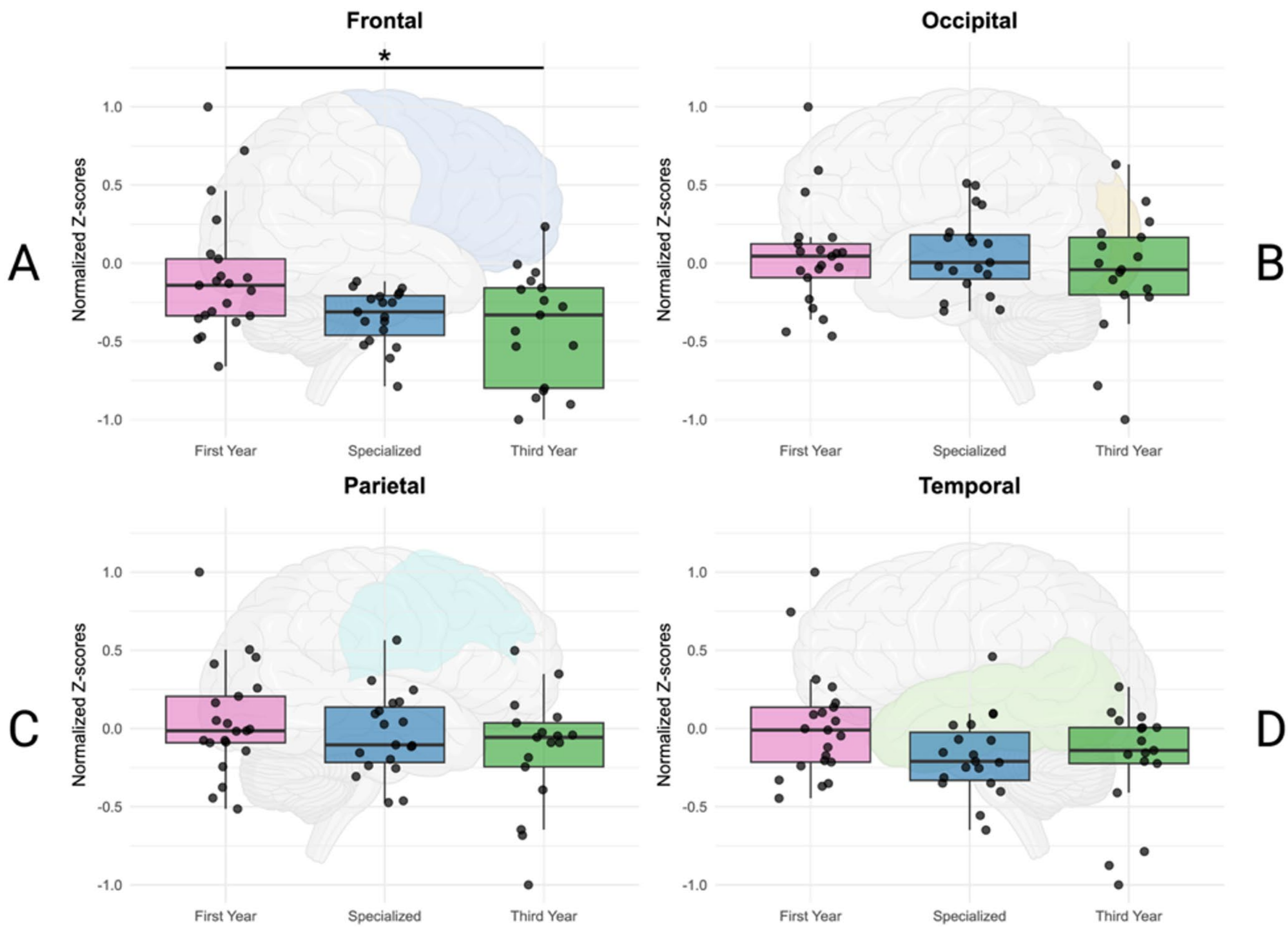

Fig. 3 Cross comparisons of beta-band lobe activity between cadet cohorts. The figure depicts the average normalized z-scores for beta-band activity for each lobe (**A**: frontal, **B**: occipital, **C**: parietal, **D**: temporal) for each cohort of cadets (first year: purple bar and dots, specialized: blue bar and dots, third year: green bar and dots). Frontal Beta activation (3 A) was found to be significantly different between the three cohort groups ($p = 0.0288$)

had a negative non-significant effect. HRV also showed a positive, non-significant effect on decision-making. It should be noted that there was a decline in the model's predictive power with the inclusion of HRV, with there being a decrease of 4% in variance explained. Full model results are reported in Table 1 below.

Regarding reasoning, our first-level analysis showed a significant overall model ($p = 0.0048$, adj $R^2 = 0.12$). Similar to decision-making, beta-band activation in the frontal lobe had a significant negative effect on reasoning scores, with a beta coefficient of -3.40 ($p < 0.01$). In the second-level model, introducing EEG measures, the model weakened ($p = 0.0685$, adjusted $R^2 = 0.09$), with the frontal lobe no longer serving as a significant predictor of reasoning ($p > 0.05$). Introducing HRV strengthened the model, resulting in a significant third model ($p = 0.0241$, adj $R^2 = 0.16$); however, none of the predictors showed significance, including frontal beta (p = 0.097). Full model results can be found in Table 2 below.

For memory, all models were significant. Our first-level model found a significant negative effect of beta-band activation in the frontal lobe on memory encoding, with a coefficient of -8.03 ($p < 0.01$) and an adjusted R^2 of 0.17. The second-level model, including All Lobes Beta, showed the frontal lobe remaining significant, with a larger effect size (coefficient = -10.92, $p < 0.01$), and the model's adjusted R^2 increased to 0.17. Introducing HRV maintained a significant overall model (adj $R^2 = 0.17$), with beta-band activation in the frontal lobe showing a significant negative effect on memory performance (coefficient = -11.83, $p < 0.01$). None of our other predictor variables were significant. Full model results can be found in Table 3 below.

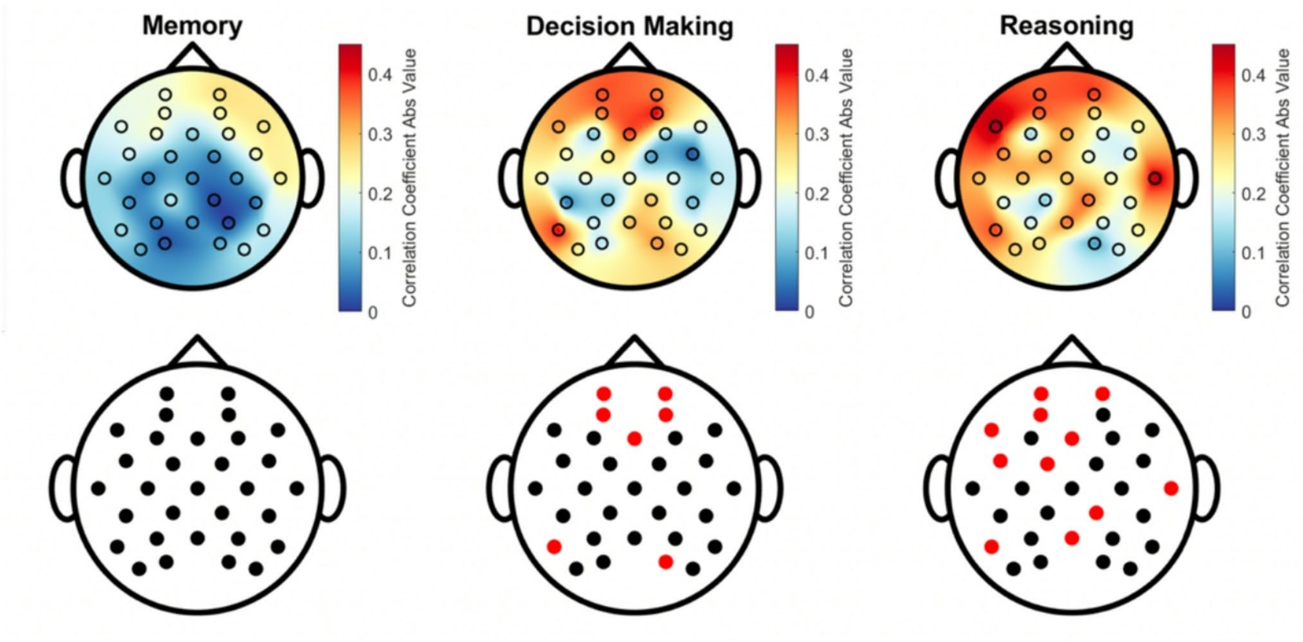

Fig. 4 Topographical maps of beta-band correlations with performance measures. Scalp distributions display the absolute value of channel-wise correlation coefficients between beta-band EEG activity and three performance outcomes: memory, decision-making, and reasoning. Bottom row: Corresponding electrode maps indicating statistically significant correlations (red dots) at individual channels (black = non-significant)

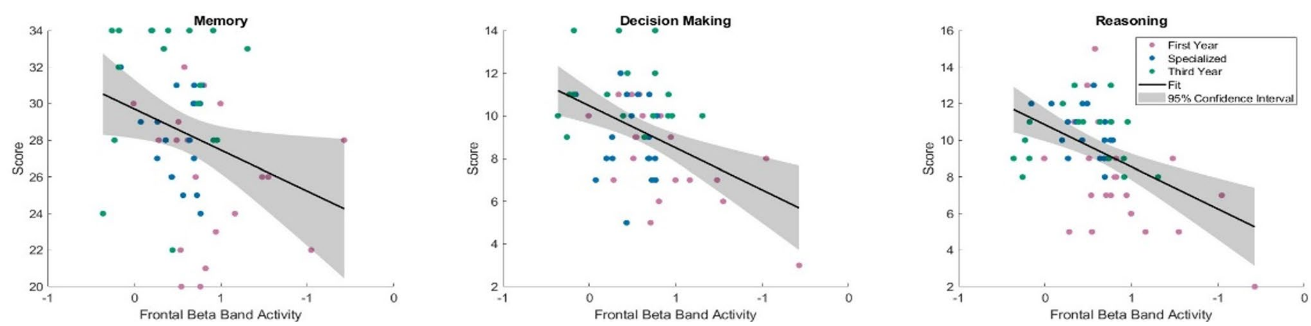

Fig. 5 Pairwise correlations between situational performance metrics and frontal beta-band activity. Scatterplots of average frontal beta-band activity plotted along the x-axis against each situational performance metric plotted on the y-axis, with a linear fit and 95% confidence interval applied to the data. These plots illustrate significant negative relationships between average frontal beta-band activity and each performance metric, with higher scores eliciting lower beta-band activity

Discussion

This research study aimed to address significant gaps in the literature on the neurophysiological correlates of situational performance in policing, focusing on a highly novel whole-brain EEG methodology and its associated analyses. Effective situational performance is critical to law enforcement, often influencing life-or-death outcomes for both officers and the public. To explore how these skills are developed and manifest in the brain, a carefully designed video-based simulation task was used to replicate stressful incidents, judgments, and decisions that officers might encounter in the line of duty. The simulation provided insights into participants' performance efficacy through several decision-making prompts, which were evaluated during a post-simulation debriefing interview. Furthermore, the study examined cadets' ability to recall key details of the incident, allowing researchers to assess the neurophysiological factors associated with memory encoding during the task itself. All situational performance metrics were significantly correlated to beta-band activation in the frontal lobes, with higher scores related to greater beta-band suppression. This novel approach to measuring, tracking, and understanding police performance skills offers valuable insights into the development and training of neurophysiological correlates of situational performance, an inherently volatile yet essential aspect of policing.

Table 1 Hierarchical linear model for decision-making

Predictors	Model 1: Beta Frontal			Model 2: All Lobes Beta			Model 3: All Lobes Beta + HRV		
	Estimates	SE	p	Estimates	SE	p	Estimates	SE	p
(Intercept)	8.75	0.32	**< 0.001*** **	8.67	0.40	**< 0.001*** **	8.59	.713	**< 0.001*** **
Frontal	-3.02	1.02	**< 0.01** **	-3.45	1.64	**< 0.05***	-5.02	1.87	**< 0.05***
Temporal				2.16	2.40	0.423	3.63	2.59	.168
Parietal				-0.64	3.65	0.871	-2.77	4.14	0.508
Occipital				-1.49	2.77	0.618	.211	3.12	0.947
HRV							0.24	1.66	0.988
p	0.0039			0.0201			0.0850		
R^2 adj	0.13			0.14			0.10		
F statistic	9.08			3.21			2.09		
N	54			54			50		

Multivariate models predicting decision-making among cadets at kuwait's National Police academy using beta band activation across brain lobes (frontal, temporal, parietal, and occipital) and ECG. models are divided based on inclusion of: (1) frontal lobe, (2) all lobes beta, and (3) EEG/ECG

The table reports coefficients marked with asterisks to denote statistical significance

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 2 Hierarchical linear model for reasoning

Predictors	Model 1: Beta Frontal			Model 2: All Beta Lobes (No HRV)			Model 3: All Beta Lobes + HRV		
	Estimates	SE	p	Estimates	SE	p	Estimates	SE	p
(Intercept)	9.00	0.35	**< 0.001*** **	9.00	0.45	**< 0.001*** **	9.14	.809	**< 0.001*** **
Frontal	-3.40	1.15	**< 0.01** **	-2.48	1.90	0.197	-4.09	2.42	< 0.097
Temporal				-2.62	2.78	0.350	-2.81	3.19	0.382
Parietal				0.81	4.07	0.842	1.33	4.19	0.752
Occipital				1.03	3.08	0.739	1.73	3.05	0.574
HRV							-1.82	1.75	0.304
p	0.0048			0.0685			0.0241		
R^2 adjusted	0.12			0.09			0.16		
F stat	8.66			2.33			2.89		
Observations	55			55			51		

Multivariate models predicting reasoning among cadets at kuwait's National Police academy using beta band activation across brain lobes (frontal, temporal, parietal, and occipital) and ECG. models are divided based on the inclusion of: (1) frontal lobe, (2) all lobes beta, and (3) EEG/ECG

The table reports coefficients marked with asterisks to denote statistical significance

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Table 3 Hierarchical linear model for memory recall

Predictors	Model 1: Beta Frontal			Model 2: All Lobes			Model 3: All Lobes Beta + HRV		
	Estimates	SE	p	Estimates	SE	p	Estimates	SE	p
(Intercept)	26.67	0.69	**< 0.001*** **	28.19	0.75	**< 0.001*** **	26.80	1.41	**< 0.001*** **
Frontal	-8.03	2.32	**< 0.01** **	-10.92	3.36	**< 0.01** **	-11.83	3.45	**< 0.01** **
Temporal				2.50	4.97	0.617	2.20	5.00	0.661
Parietal				5.60	7.71	0.471	4.58	7.90	0.565
Occipital				0.60	5.80	0.919	1.12	5.86	0.849
HRV							2.23	3.21	0.470
p	0.0011			0.0091			0.0163		
R^2 adjusted	0.17			0.17			0.17		
F stat	11.94			3.38			3.11		
N	53			53			54		

Multivariate models predicting performance memory recall among cadets at kuwait's National Police academy using beta band activation across brain lobes (frontal, temporal, parietal, and occipital) and ECG. models are divided based on inclusion of: (1) frontal lobe, (2) all lobes beta, and (3) EEG/ECG

The table reports coefficients marked with asterisks to denote statistical significance

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Neurophysiological Correlates of Situational Performance

Previous studies and reviews have highlighted the potential of neurophysiological markers in predicting situational performance in policing (Andersen et al., 2024; Anderson et al., 2019; Chan et al., 2022). However, these studies have often been limited to self-report measures or easily accessible methods for assessing cardiovascular physiology, primarily through HR data collected via ECG. This study aimed to incorporate neurological activity recorded through EEG, which has been scarcely used in policing contexts (Alexander et al., 2024; Johnson et al., 2014; Muñoz et al., 2020). Notably, none of these precedent studies have matched this study's sample size ($n=58$ versus 27, 24, and 10 participants, respectively), simulation methodology, or situational performance outcomes. The simulations employed by past research focused on lethal force 'shoot/no-shoot' decision-making. In contrast, this study assessed broader procedural aspects of police performance during crisis calls, ranging from routine procedures (e.g., requests for identification) to sense-making of a domestic dispute involving multiple persons in crisis, as well as potential use of force and apprehension decisions. Additionally, this study serves as the first to explore both EEG and ECG metrics in police cadets at different stages of a prolonged academy curriculum, allowing for a comprehensive assessment of the development and predictive significance of neurophysiological mechanisms underlying essential policing skills related to situational performance.

Preliminary analyses revealed significantly strong bivariate correlations between cadets' beta-band activity and all three situational performance metrics, especially in the frontal cortex (Fig. 4). Further testing confirmed that frontal lobe beta-band activity retained significance when other correlates were introduced, including HRV (Table 1 and 3). The results partially confirmed the first hypothesis regarding the frontal lobe's significance in predicting situational performance elements. These findings align with prior literature emphasizing the central role of the frontal cortex in higher-order cognitive tasks (e.g., executive functioning and attentional focus) (Friedman & Robbins, 2022). These findings are corroborated by research across various domains of human performance, including memory recall (Foy & Foy, 2020), emotional regulation (Knyazev, 2007), and motor skill acquisition (Mak et al., 2013). Qiu et al. (2018) investigated neural mechanisms underlying metacognition and decision-making, finding that increased activation did not necessarily correlate with improved decision confidence, suggesting that lower activation facilitates clearer decision-making. Likewise, Liu et al.'s (2023) study on regional gray matter volume in the posterior cingulate cortex found that lower activation correlated with higher levels of reflective thinking and self-regulation during decision-making tasks, further suggesting that decision-making efficacy is linked to efficiency in neural resources. Consequently, it can be concluded that cadets who performed better in the simulation task were also more efficient in their neurophysiological processing of the critical incident, requiring less cognitive load and beta-band activation in assessing and reacting to simulation segments than those who scored worse. This study is the first to observe such a relationship between neurophysiological activity and situational performance within the fields of criminal justice and applied policing.

Frontal beta-band suppression was also found to be a consistent correlate of improved memory encoding and recall (Fig. 5; Table 3). Literature has elaborated on the nuances of memory recall and encoding during stressful incidents, suggesting that impairments typically observed in memory encoding do not occur under certain conditions, as supported by meta-analytic work (Shields et al., 2017). These conditions include elements all found within the current study task: the encoded material was task-relevant and stressor-related, there was a short time interval between encoding and retrieval (approximately 20 min), and central details (rather than peripheral details) were tested. While HRV was not a significant correlate of memory, previous research suggests that cadets who experienced greater subjective self-reported stress and/or greater increases in cortisol may have experienced an impairing effect on memory recall (Shields et al., 2017; Wolf, 2017), an idea that should be further explored in future studies. The frontal lobe, and particularly the prefrontal cortex, plays an important role in memory encoding and retrieval (Fletcher & Henson, 2001). In the current study, we found a significant negative effect of frontal lobe beta-band activation on memory performance. Similarly, Foy and Foy (2020) found alpha, beta, and gamma activity to be inversely correlated with performance outcomes during memory tests. Together, these findings suggest that individuals with better memory recall during cognitive tasks exhibit lower activity in these frequency bands, reflecting efficiency in the use of neural resources to facilitate improved cognitive performance.

Lastly, HRV played a minimal role in predicting performance, and none of the situational performance metrics showed significant associations with HRV, even during preliminary pairwise analysis. Various cardiovascular measures recorded by ECG have been widely used to conceptualize stress and arousal in policing studies—particularly in those examining situational stress and its impact on performance (Baldwin et al., 2019; Bennell et al., 2021; Di Nota & Huhta, 2019). As recently stated by Andersen et al. (2024), it is important that the interpretation of physiological biomarkers in applied police contexts take into consideration the basic theory and methodological limitations behind them. Although HRV did not exhibit statistical significance as a predictor, incorporating it into multivariate analyses enhanced one of our reasoning model's significance and ability to explain variance in cadets' critical thinking abilities. This suggests that while HRV may not directly predict performance, it could contribute to a more comprehensive understanding when considered alongside other neurophysiological variables. These findings address a gap in the literature, as no prior studies have examined HRV and EEG in tandem when assessing their predictive power in multivariate models for police-related studies. This research offers critical implications for future studies aiming to develop effective neurophysiological methods for predicting police performance, including cautious interpretation and appropriate use of highly specific biomarkers in highly specific police contexts. In so doing, future applied research can avoid the potential misinterpretation and misuse of findings to inform evidence-based police policies and practices that can have significant real-world consequences.

Notably, inclusion of posterior regions did not improve model performance beyond what was already accounted for by frontal electrodes. These findings suggest that frontal beta activity alone may be sufficient to capture meaningful variance in task performance, offering both theoretical and practical implications. EEG activity is generally highly correlated across the entire cortex. As illustrated in Fig. 5, correlation maps across memory, decision asking, and reasoning tasks reveal that while beta-band activity is distributed across multiple cortical areas, the strongest and most consistent effects are localized to the frontal cortex. This widespread activation pattern supports the idea that multiple regions are engaged during complex cognitive processing; however, in our multivariate models, only frontal activity significantly predicted performance outcomes. This suggests that while other regions are co-active, their contribution is likely redundant due to the high spatial correlation inherent in EEG signals. These findings reinforce the functional primacy of the frontal cortex in tasks involving executive control and complex action planning. Therefore, our model could not capture any information in the other lobes beyond what the frontal lobe is providing related to performance. From a translational perspective, this supports the feasibility of using low-cost EEG systems with limited frontal coverage in applied settings such as police training, performance monitoring, or neurofeedback interventions.

Expertise, Knowledge, and Training

Preliminary cross-comparison testing revealed significant differences in performance between cadets based on cohort, with third-year cadets who had completed 2.5 years of training at the time of data collection scoring significantly higher in decision-making and memory recall compared to first-year cadets with 4 months of training, and performing comparably in reasoning to specialized cadets that had completed 6 months of academy training and held post-secondary degrees. Neural activity followed a similar pattern, with first-year cadets showing the highest beta-band activity (M = -0.061), followed by specialized cadets (M = -0.224), and third-year cadets (M = -0.289) as seen in Fig. 3. Cadets at Kuwait's Saad Al-Abdulla Police Academy undergo rigorous training, ultimately graduating as lieutenants in Kuwait's National Police Force after a four-year program. This training not only prepares them physically but also immerses them in legal knowledge and skills essential for their future roles in law enforcement. These skills are progressively refined over their time at the academy, establishing a knowledge base that supports officers during high-stress situations. Thus, it is expected that third-year cadets would be more knowledgeable than the other cadet groups, having trained at least 22 months more than both of the other cohorts. Indeed, this assumption was confirmed by analyses of both performance and neurophysiological measures, with third-year cadets scoring highest in two out of three performance metrics and displaying the lowest beta-band activity across all three cohorts during the simulation task (Fig. 3).

These results contrast with previous studies examining neurophysiological activity among experts and novices engaging in simulation tasks. In policing contexts, Alexander et al. (2024) found that expert officers exhibited increased frontal midline theta power compared to novices during shoot/no-shoot exercises, attributing this to their enhanced attentional control and threat preparedness. They also attributed shorter beta rebound periods to experts' training and experience with swiftly recovering after executing an action. Muñoz et al. (2020) found police officers' frontal theta/beta ratios correlated significantly with task difficulty; specifically, more challenging simulated shooting trials were associated with higher activation in both frequency bands. This suggests simulation segments used in this study were more challenging for lesser-trained cadets, leading to elevated beta-band activity across All Lobes Beta relative to third-year cadets. This heightened activity could reflect increased cognitive load and mental effort with more limited training and procedural knowledge. Johnson et al.' (2014) findings corroborate these assumptions, showing expert officers to show lower engagement with stimuli during critical decision-making, signifying their more refined and efficient assessment of critical procedural cues when compared to novices. Similar to the current study task, the additional training afforded to third-year cadets could elicit learning-induced efficiency in frontal beta-band activation while observing a complex action sequence that is part of their embodied motor repertoire. In line with our HRV implications, it is important to note that comparing findings between these EEG studies, which are the closest in theoretical, methodological, and analytical approaches, is still very difficult given differences in each study's chosen brain/electrode regions, frequency bands, and signal processing techniques (e.g., log or z transformed, baseline corrected). Further investigations are needed to establish standards for applied psychophysiological studies on motor learning and action processing (i.e., observation, visualization/imagery, and execution).

Study Limitations and Future Directions

Our research addresses important gaps in understanding the relationship between neurophysiological mechanisms and situational performance in policing. However, several limitations exist regarding the generalizability and methodological approaches employed. One limitation concerns the video simulation methodology used for data collection. Although computer- and video-based simulations are common tools for examining situational awareness and police performance (Di Nota et al., 2023; Stenshol et al., 2023), these methods present inherent limitations in realism and ecological validity compared to real-worldpolicingscenarios (Kleygrewe et al., 2024). This was most relevant during preliminary analysis on cadets' HRV, which showed no significant difference between the baseline and stimulus phases, unlike other similar studies on officer situational performance. Advanced technologies such as virtual reality (VR) enhance immersion (Caserman et al., 2018, 2022; Zechner et al., 2023), but practical constraints related to EEG interference and limited collection time prevented their use in our study.

Another methodological consideration involves the aggregation of EEG and ECG data across the simulation duration, averaging brain activity into the four major lobes. Although this exploratory approach provided novel insights and allowed us to identify the relative predictive power of EEG and ECG on performance, future studies should investigate EEG dynamics more granularly. Potential areas include hemispheric asymmetry, frequency-band coupling, anticipatory stress versus task activation, and recovery periods (see Muñoz et al., 2020). Additionally, relatively affordable EEG equipment and research collaborations can address funding and resource limitations for neurophysiological assessments in policing.

A further methodological consideration involves our approach to assessing situational performance, focusing primarily on less visible cognitive skills (e.g., reasoning, memory) rather than observable behaviors or objective measures such as shoot/no-shoot decisions and eye-tracking (Huhta et al., 2022; Nieuwenhuys & Oudejans, 2010). This approach, while valuable, introduces potential response biases due to post-task evaluations. Incorporating live response metrics during decision prompts could improve assessments of real-time decision-making accuracy and response times. The uniformity of our professionally produced simulation, due to resource constraints, limited scenario variability and response diversity but helped maintain consistent evaluation standards, minimized reliability issues, and provided logistical feasibility.

Promising research suggests integrating real-time electrophysiological biofeedback in VR scenarios could enhance officer performance during stress (Michela et al., 2022). The identification of frontal-beta activity as a significant predictor of situational performance has important implications for neurofeedback applications in real-world training contexts. Beta band activity could serve as a biomarker to train individuals to self-regulate during decision-making tasks under pressure. In addition, the fact that the frontal lobe emerged as a significant correlate for performance could motivate the use of low-cost EEG systems with limited coverage. Further research should be done to investigate if Frontal Beta could indeed be used as a biomarker to improve performance in a variety of settings. However, in this current study, focus was mainly centered on decision-making tasks that require complex situational analysis, where the officers must process ambiguous and evolving information and must make the appropriate judgment while adhering to protocols and regulating their own emotional and physiological arousal. Therefore, one must note that our findings need not translate to other tasks that are more reactionary and straightforward, such as a shoot/no shoot task. Future studies should further refine methods for integrating real-time neurophysiological and behavioral measures, promoting more comprehensive assessments of police performance under realistic conditions.

Conclusion

Law enforcement officers worldwide frequently encounter unpredictable and life-threatening situations, and are expected to respond effectively to maintain public safety. In response to highly publicized critical incidents, researchers have sought to identify correlates of situational outcomes, often incorporating neurophysiological and psychological elements associated with them. However, there is a notable gap in applied police research regarding advancements in neurophysiological data collection, with studies commonly using simple peripheral measures to assess stress physiology (mainly ECG) in the context of situational performance in policing. This study provides substantial advancements in the exploration of neurophysiological markers underlying decision-making, reasoning, and memory of simulated critical incidents. The findings provided novel insight into the predictive capacity of beta-band activity on performance efficacy in the frontal cortex - the area of the brain tasked with executive functions and emotional control, both of which are essential during high-stress scenarios encountered in policing. Furthermore, multivariate analyses downplayed the significance of HRV in comparison to EEG on performance outcomes. Future applied research should incorporate EEG measures to further understanding of the neurophysiological mechanisms supporting numerous aspects of police performance and training.

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Data Availability Requests for access to the study data should be directed to the corresponding author. Sharing data is subject to the approval of Kuwait's Saad AlAbdulla Police Academy.

Declarations

Conflict of interest The authors declare no competing interests.

Ethical Approval This study received full ethical approval from Kuwait's Saad AlAbdulla Police Academy and Kuwait University's Institutional Review Board (KU-ENG-14-11-23). All participating cadets provided written consent before participation.

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